1. The 30-second pitch

We predict whether a credit-card client will have **payment difficulty next month**. We trained three interpretable ("glass-box") models — a **decision tree**, **logistic regression (LR)** and an **Explainable Boosting Machine (EBM)** — under *one* protocol (same split, same 5 CV folds, same thresholds logic, same 4 local cases). The **EBM wins** (CV AUC 0.776 vs 0.761 LR vs 0.752 tree, in 5/5 folds) and even beats a black-box gradient-boosting model (0.768): *interpretability cost nothing here*. All models agree that **recent repayment behaviour** drives risk, but they disagree on *which* correlated variable carries the signal and on the *shape* of the effect — and that explains the clients where they disagree. We submitted the EBM.

2. Data and problem

- 25,000 labelled + 5,000 unlabelled (`test.csv`) clients; 15 predictors; no missing values. Based on *Default of Credit Card Clients*, with a semi-synthetic target.
- Predictors: credit limit; sex, education, marital status, age; 8 six-month summaries (latest repayment status, max delay, delayed months, average bill, bill change, total payment, payment-to-bill ratio, zero-payment months); September utilization; repayment-assistance plan.
- **Imbalanced**: only 21.2 % have difficulty. Always saying "no" already gives **78.8 % accuracy** ⇒ accuracy alone is misleading.
- **Repayment status codes**: ≥ 1 = months behind; $-2, -1, 0$ are undocumented "not behind" codes (the data dictionary warns not to invent an order for them).

3. Basic concepts (glossary)

Evaluation

- **Risk score / probability** $\hat{p}$: model output in $[0, 1]$.
- **Threshold** t: flag the client if $\hat{p} \geq t$. Turns a score into a yes/no decision.
- **Confusion matrix**: TP (difficulty caught), FN (missed), FP (false alarm), TN (correctly not flagged).
- **Accuracy** = (TP+TN)/n: share of all clients right.
- **Precision** = TP/(TP+FP): of the flagged clients, how many really had difficulty (few false alarms).
- **Recall / sensitivity** = TP/(TP+FN): of the clients in difficulty, how many we caught (few misses). *Specificity* = recall of class 0.
- **F1** = harmonic mean of precision and recall; good single number for imbalanced data.
- **Balanced accuracy (BA)** = mean of recall of class 1 and class 0. Chance = 0.5 regardless of imbalance.
- **ROC curve**: true-positive rate vs false-positive rate for *every* threshold.
- **AUC**: area under ROC. = probability that a random client *with* difficulty gets a higher score than a random client *without*. 0.5 = guessing, 1 = perfect ranking. **Threshold-free**, so we used it to compare and tune models.
- **Calibration**: $\hat{p} = 0.3$ really means ≈30 % observed difficulty.
- **Cohen's** κ: agreement between two models corrected for agreement by chance.

Validation

- **Train/validation split**: 80/20 (20k/5k), *stratified* (same 21 % rate in both), seed 411.
- **k-fold cross-validation (CV)**: split training data into 5 folds; train on 4, evaluate on 1, rotate. Gives a mean ± std. **Stratified**: each fold keeps the class rate.
- **Shared folds / paired comparison**: all models use the *same* 5 folds, so we compare them fold by fold ("EBM wins 5/5 folds").
- **Out-of-fold prediction**: each training client's score from the model that did *not* see it. Used to choose thresholds without cheating.
- **One-standard-error rule**: among settings whose CV AUC is within 1 SE of the best, take the *simplest*. Differences smaller than the noise are not real, so prefer the smaller model.
- **Hyperparameter**: setting chosen before training (depth, C, bins…), tuned by CV (`GridSearchCV`).
- **Data leakage**: using information you shouldn't (e.g. test data). Test set never used for fitting or selection; validation used *once* at the end.

Modelling and interpretation

- **Glass box vs black box**: a glass-box model can be read directly (rules, coefficients, curves); a black box (e.g. gradient boosting) needs post-hoc explanation.
- **Global explanation**: how the model behaves overall (which variables matter, shape of effects). **Local explanation**: why *this* client got *this* prediction.
- **Odds** $= p/(1-p)$. **Log-odds (logit)** = log odds. Additive models (LR, EBM) add up contributions on the log-odds scale; $p = 1/(1 + e^{-\text{log-odds}})$ (sigmoid). **Odds ratio** $= e^{\beta}$.
- **Standardization**: $(x - \text{mean})/\text{SD}$, so LR coefficients are comparable ("per SD").
- **One-hot / drop-first (reference) coding**: one 0/1 column per category; drop one category to avoid perfect collinearity with the intercept.
- **Regularization (L2)**: penalizes large coefficients; in sklearn $C = 1/\text{strength}$ (smaller C = stronger shrinkage).
- **Collinearity**: predictors that carry the same information; coefficients become *conditional* and can flip sign.
- **GAM (generalized additive model)**: logit $= \beta_0 + \sum_j f_j(x_j)$, one curve per variable.
- **Impurity (Gini)**: how mixed the classes are in a node; a tree split is chosen to reduce it most.
- **Rashomon effect**: many models fit equally well while relying on different variables ⇒ importance is a property of the model, not of the data.
- **Simulatability**: a human can compute the prediction by hand (the tree's rules).
- **Causality**: all our explanations are *associations*, not causes.

4. How it was done (pipeline)

1. Explore data: imbalance, skewed money variables, undocumented status codes.
2. Split once 80/20, stratified. Define the same 5 stratified folds on the training set.
3. For each model family, grid-search hyperparameters by **CV AUC** (preprocessing is inside the pipeline, so it is learned on the training folds only); apply the one-SE rule for tree and LR.
4. Compute out-of-fold probabilities, choose thresholds t_{acc} and t_{bal} on the grid 0.05–0.95.
5. Add references: majority class (chance) and untuned `HistGradientBoosting` (black box).
6. Confirm everything once on the 5,000 validation clients (all within ≈1 fold SD of CV).
7. Interpret each model globally and locally with its own explanations (InterpretML views); compare importances (normalized to shares) and 4 local clients.
8. Subgroup audit by recorded sex (descriptive).
9. Refit the EBM on all 25,000 labelled rows, predict `test.csv` at $t_{\text{acc}} = 0.49$ (flags 10 % of test clients).

5. What was considered (decisions and why)

- **AUC as main metric**: imbalance makes accuracy misleading; AUC is threshold-free.
- **Two thresholds**:
  - t_{acc} (0.47–0.49, maximizes accuracy) → used for the **submission**, because the test set is graded on accuracy. Flags ≈10 %, catches ≈30 % of difficulties.
  - t_{bal} (0.18–0.23, maximizes balanced accuracy) → our **analysis point** (misses and false alarms both count). Flags ≈30 %, catches ≈62 %.
  - t_{bal} ≈ base rate (21 %): for a calibrated model, BA is maximized by flagging anyone riskier than average.
- **Thresholds instead of class weights**: (1) one mechanism for all three families (weights change the tree/EBM structure differently); (2) probabilities stay calibrated, so local explanations describe the real 21 % risk; (3) one fitted model serves both goals — only t changes.
- **One-SE rule**: in an interpretability project, prefer the smaller model if the data can't tell them apart (tree: 17 leaves instead of 32; LR: $C = 0.01$ instead of 0.1).
- **Log transform for LR** chosen *inside* the CV grid (+0.012 AUC): money variables are very right-skewed. Signed $\log(1+|x|)$ because bills can be negative.
- **Tree keeps raw units**: splits don't change under monotone transforms, raw thresholds are readable.
- **EBM gets raw columns**, `interactions=0`: exactly one curve per original variable (pure GAM).
- **Same 4 local clients** for all models, chosen by fixed rules (not cherry-picked).
- **Black-box reference** to measure the "price" of interpretability.
- **Fairness**: descriptive check by sex; **causality**: assistance plan is a marker, not a cause.

6. Results at a glance

5-fold CV	AUC	Acc [t_{acc}]	BA [t_{bal}]	F1 [t_{bal}]
Tree (17 leaves)	.752	.812	.692	.493
LR (C=0.01, log)	.761	.814	.695	.501
EBM	**.776**	**.817**	**.705**	**.517**
Majority class	.500	.788	.500	—
Black box (HGB)	.768	.815	.701	—

- EBM − LR = +0.014, EBM − tree = +0.023 AUC, each in 5/5 folds; LR beats tree 5/5.
- Validation (5k): AUC tree .749, LR .766, EBM .775. Matches CV.
- Models make the **same decision for 87 %** of validation clients (κ 0.76–0.82).
- Thresholds: tree .49/.18, LR .47/.22, EBM .49/.23 (t_{acc}/t_{bal}).

7. Decision tree (CART)

How it works: repeatedly splits clients with yes/no questions ("status ≤ 0.5?") choosing the split that most reduces impurity; a leaf's risk = share of difficulty among its training clients.

Setup: `max_depth` 2–10, `min_samples_leaf` 25–400. AUC plateaus at depth 5. Best: depth 5, leaf ≥25, 32 leaves, AUC .755. One-SE: depth 5, leaf ≥400, **17 leaves**, AUC .752.

How to read the tree figure

Each box: **question** (yes → left/True, no → right/False); **samples** = % of training clients reaching the node; **value** = [share no difficulty, share difficulty] (2nd number = risk); **class** = majority (yes only if risk > 0.5); colour orange = no, blue = yes, darker = purer; grey (...) = truncated branches. `status <= 0.5` means status ≤ 0 (not behind). `plan = no <= 0.5` true means the client *has* a plan. *Root*: 100 %, [0.788, 0.212] = base rate. Behind on payments (status ≥1): 22.8 % of clients, 43.8 % risk. Then ratio ≤ 0.095: 15.3 %, risk 0.538 → rule R1. **Caution**: the figure's "class" uses 0.5, but we decide at $t_{bal} = 0.18$, so leaves with risk 0.26–0.43 are also flagged.

Global interpretation: 6 rules

Merging sibling leaves with the same class: 17 leaves → 9 rules, **6 predict difficulty** ("flag if any rule fires"):

- **R1** status ≥1 & ratio ≤0.095: risk .54, 15.3 % of clients, 39 % of all difficulties.
- **R2** status ≤0 & ratio ≤0.025: .43.
- **R3** status ≥1, ratio >0.095, zero-pay months ≥4: .33.
- **R4** status ≤0, ratio 0.025–0.105, age ≤24: .32.
- **R5** status ≥1, ratio 0.095–0.315, zero-pay ≤3: .28.
- **R6** as R4 but age ≥25 and plan = yes: .26.

Uses only 7 of 15 variables; status + ratio = 85 % of importance (impurity importance). Ignores utilization, credit limit, bills, total payment, max delay, sex, education, marital status.

Local explanation = the path to the client's leaf. **Pros**: simulatable, easy to communicate. **Cons**: coarse (only 17 probability levels, ROC is piecewise linear), hard thresholds, lowest AUC.

8. Logistic regression

How it works: log-odds = $\beta_0 + \sum_j \beta_j x_j$; probability via sigmoid. Linear (straight line) in each variable on the log-odds scale.

Setup: L2, `lbfgs`; numeric variables standardized; categorical drop-first (references: female, graduate school, married, no plan); signed log of 4 money variables; $C \in \{0.001..100\}$; one-SE → $C = 0.01$ (AUC .761).

Global: coefficient = change in log-odds per 1 SD (or vs reference category). Largest: total payment −0.51; utilization +0.46; plan +0.44 (odds ratio 1.55); delayed months +0.38; latest status +0.37. Coefficients stable across the 5 folds.

Caution — conditional coefficients (collinearity), sign opposite to the marginal correlation:

- max delay: corr +0.27, coef −0.08 (corr 0.83 with delayed months).
- credit limit: corr −0.12, coef +0.11 (given utilization and bills).
- payment ratio: corr −0.14, coef +0.10 (given total payment).

Read: "holding the others fixed", not "the effect of". **Linearity**: forces status codes $-2, -1, 0$ onto a line; cannot represent U-shapes.

Local explanation: contribution = coefficient × standardized value; intercept + sum = log-odds (intercept −1.63).

9. EBM (Explainable Boosting Machine)

How it works: a GAM, logit = $\beta_0 + \sum_j f_j(x_j)$, learned by *boosting*: many rounds, each fitting a tiny tree on *one* variable at a time (round-robin) with a small learning rate. The trees of each variable are summed into its **shape function** f_j (a step curve). Several bagged fits give the uncertainty bands. Accuracy close to boosting, but each variable's effect is one curve you can plot.

Setup: InterpretML, `interactions=0`, raw columns (bins numeric variables itself, categories as nominal). Grid `max_bins` {32,256,1024}, `min_samples_leaf` {4,50}: AUC only .7746–.7757 (robust). Chose 256 bins, leaf 50.

Global (shape functions; log-odds, centred at 0):

- **Payment ratio, U-shaped** (most important): paying ≈0 → up to +0.7; lowest risk at 0.5–1.2; above 1.7 back to +0.4 (catch-up payments, tiny/negative bills; 2 % of clients).
- **Repayment status**: flat over $-2, -1, 0$; rises in steps from 1 (no invented order).
- **Zero-payment months**: 1–2 *protective*, 3+ harmful, +0.84 at 6. Tested with `monotonize` inside CV: AUC drops .7756

→ .7728 ⇒ the dip is real.
- **Age, U-shaped**: <25 +0.58, >57 ≈ +0.4; 30–55 slightly protective.
- **Utilization**: rises to 1.0, jumps above ≈1.4 (over-limit, 49 % rate).
- **Assistance plan** +0.41 (no plan −0.11).

Bands narrow where data are dense, wide in sparse tails → read jumps there with caution. **Importance** = mean absolute contribution. **Local explanation**: value of each f_j for the client; intercept (−1.58) + sum = exact log-odds.

10. Global comparison

Importances normalized to shares of each model's total (units differ).

- **Agree**: recent repayment behaviour (delayed months, latest status, plan in every top 5). Tree and EBM almost same ranking (rank corr 0.86).
- **Differ**: LR puts total payment (16.6 %) and utilization (15.1 %) first, ratio only 10th (3.4 %); ratio is 1st for EBM (19 %), 2nd for tree (33.6 %). Same information (ratio = payment/bill), different carrier: a linear model takes it from log total payment; flexible models use the ratio ⇒ **Rashomon effect**.
- Tree concentrates (85 % on 2 variables); LR and EBM spread importance.

11. Local cases (validation, decision at t_{bal})

Chosen by fixed rules: A, B = most typical correct client of each class; C, D = largest disagreement.

- **A** (no difficulty, all right; .12/.13/.10): age 23 is the only risk; EBM +0.56 for age vs LR only +0.13 (linear in age).
- **B** (difficulty, all right; .51/.47/.53): status 2 is the main driver (+0.66 LR, +0.72 EBM) + plan ≈ +0.4. Sits on the boundary → why recall is limited.
- **C** (difficulty; tree .15 *missed*, LR .50, EBM .55 flagged): negative bill, ratio 4.39. Tree reads high ratio as "pays well"; EBM's U-shape reads it as irregular account (+0.48) + 4 delayed months. *Additive models right.*
- **D** (no difficulty; tree .06 right, LR .39 / EBM .28 *false alarm*): up to date but utilization 2.0 (+1.80 LR, +1.34 EBM). Tree ignores utilization. EBM's over-limit jump learned from only 0.7 % of clients. *Tree right.*

Lesson: conflicts reveal each model's inductive bias (hard thresholds, linearity, freedom in sparse regions); a local explanation is only as trustworthy as the global shape behind it.

12. Fairness and causality

- Sex: observed rates 20.5 % (F) vs 22.3 % (M). Gaps in flag rate ≤0.9 pts, TPR 1.8–2.1 pts, FPR 0.2–0.8 pts. Sex unused by tree, 2.4 % LR, 1.3 % EBM (least important). *Descriptive only*: similar error rates don't prove fairness.
- Assistance plan ≈ +0.4 in all models: plans are given *after* the lender sees stress ⇒ marker of prior difficulty, not harmful. Must not be used to deny support.

13. Likely questions

Q: Why not just accuracy?

A: 78.8 % is free (always "no"). AUC measures ranking quality independent of the threshold; BA and F1 account for both classes.

Q: Why is the EBM better than the black box?

A: The data are tabular with meaningful features and mostly additive effects; the EBM captures the non-linear shapes. The black box was untuned, so a tuned one might gain a little — but the gap would have to justify losing inspectability.

Q: Why thresholds and not class weights?

A: One mechanism for all models, calibrated probabilities (explanations in real risk), one model for both goals. In LR weights ≈ shifting the intercept anyway.

Q: Why 17 leaves and not 32?

A: One-SE rule: AUC .752 vs .755 is within noise; half the leaves.

Q: Why is t_{bal} so low (0.18–0.23)?

A: For a calibrated model, BA is maximized by flagging clients above the base rate (21 %).

Q: Why does max delay have a negative coefficient?

A: Collinearity: it is 0.83-correlated with delayed months; the coefficient is conditional on the others. Not a protective effect.

Q: How do you compute an LR local explanation?

A: Coefficient × standardized value per variable; intercept + sum = log-odds; sigmoid gives p.

Q: Why `interactions=0`?

A: One curve per variable, fully readable. Possible improvement: test a few pairs (e.g. status × ratio) to see if additivity costs accuracy.

Q: Is the zero-payment dip an artefact?

A: We tested it: forcing monotonicity lowers CV AUC (.7756 → .7728), so we kept the learned shape.

Q: Why do the models disagree on important variables?

A: Rashomon effect: correlated predictors carry the same information; each model picks a different carrier.

Q: Any data leakage?

A: Test never used; all choices by CV on training; validation touched once; preprocessing fitted inside the CV pipeline.

Q: Which model would you deploy?

A: EBM for prediction and explanation; the tree's rule set as communication device and sanity check.

Q: How good is the model in absolute terms?

A: Moderate: ≈60 % of difficulties caught at ≈21 % false-positive rate — limited by the signal in six months of behaviour, not by the model family.

Q: How were the test predictions made?

A: EBM refit on all 25k labelled rows, $t_{\text{acc}} = 0.49$; flags 10 %; agrees with the other two models on 96–97 %.

14. Limitations and improvements

- Rule set has near-duplicates (R4/R6, R3/R5) → rule-list learner (CORELS, `imodels`).
- EBM with selected interactions; edit shapes only in sparse tails, checked by CV.
- LR with bins/splines for ratio, status, zero-pay months.
- Nested CV to remove the small optimism of tuning + thresholds on the same folds (validation suggests it is negligible).
- Cost-sensitive thresholds agreed with domain experts; fuller fairness analysis.